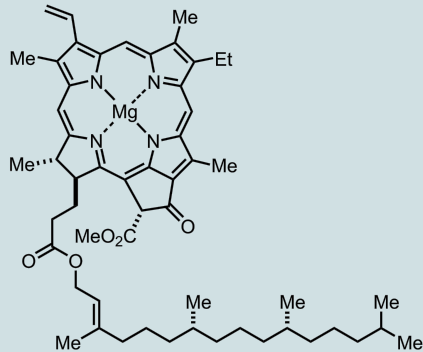


Modeling Freshwater Health: A Machine Learning Approach to Chlorophyll-A Prediction

IST 707: Applied Machine Learning

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Chlorophyll-A (CHLA): What & Why?

Introduction

Freshwater ecosystems are critical to biodiversity, public health, and recreational value, and their primary productivity-often measured through **chlorophyll-a (CHLA) concentrations- serves as a key indicator of ecological health**. High chlorophyll-a levels often indicate excessive algal growth, which can lead to harmful algal blooms (HABs) that deplete oxygen, block sunlight, produce toxins, and severely harm aquatic life and drinking water quality.

Traditional prediction methods are often site-specific and limited in scope, failing to capture the complex, non-linear relationships between nutrient levels and algal growth. To address this, our project leverages nationwide environmental data from the [EPA's National Aquatic Resource Surveys \(NARS\)](#) to develop a more generalizable and data-driven approach to forecasting CHLA concentrations. **This project aims to empower environmental scientists, EPA analysts, and local water managers with better tools for assessing lake health, anticipating harmful algal blooms, and guiding resource management decisions.**

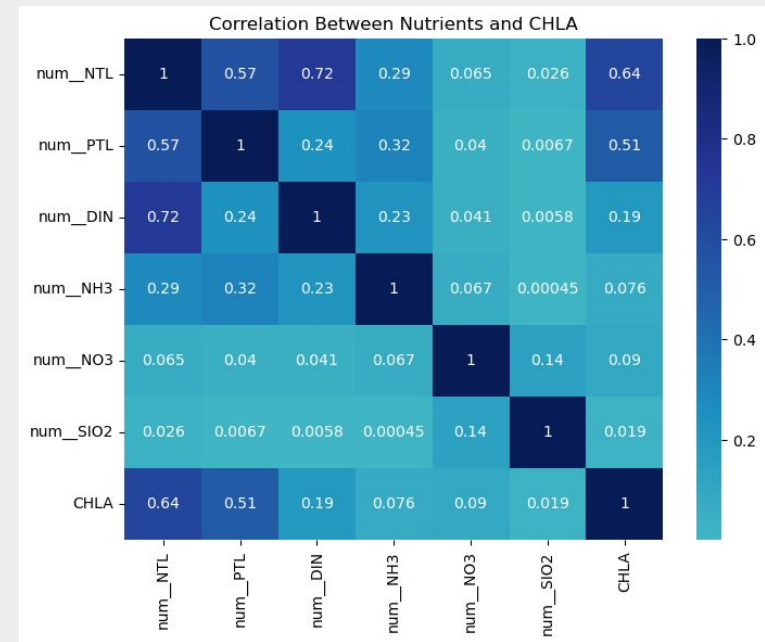
Data Source & Quality

- Data sourced from the **EPA's National Aquatic Resource Surveys (NARS)** program.
- Includes nutrient levels, CHLA, turbidity, temperature, lake type, and elevation.
- Nationwide scope with standardized sampling and quality assurance.
- Highly reliable and appropriate for large-scale predictive modeling.



The Data

- **Dataset Structure & Feature Overview:**
 - 1,082 total observations with 249 numeric features.
 - Target variable: Chlorophyll-a (CHLA), a proxy for primary productivity.
 - Features include nutrient levels, geographic attributes, and binary indicators.
- **Variance & Distribution:**
 - High variance in Silicon Dioxide (SiO₂), Dissolved Inorganic Nitrogen (DIN), and Nitrate (NO₃).
 - Dominant categorical values: Michigan and California.
- **Pre-Processing:**
 - Merged EPA datasets on common site ID and dropped rows with missing CHLA values.
 - Standardized numerical features and one-hot encoded categorical variables.



This correlation heatmap shows that chlorophyll-a (CHLA) is **most strongly associated with Total Nitrogen** (NTL, 0.64) and **Total Phosphorus** (PTL, 0.51), confirming their key role in lake productivity. Other nitrogen forms like DIN, NH3, and NO3 show weak correlations, and Silica (SiO₂) is largely irrelevant. **These results suggest that total nutrient measures are more useful for predicting CHLA**, while less-informative features could be reduced or excluded from modeling.

Model Building

We selected three supervised regression models for comparison due their strengths in handling non-linearity, interpretability, and performance:

- **Random Forest Regressor**
 - Ensemble method
 - Builds multiple decision trees and averages their outputs
 - Handles non-linearity well and provides feature importance measures
 - Hyperparameters were tuned using [GridSearchCV](#).
- **XGBoost Regressor**
 - Gradient boosting technique
 - Performance on tabular datasets
 - Additive model to correct errors from previous trees
 - Parameters (learning rate, number of estimators, and max depth) were optimized using cross-validated grid search

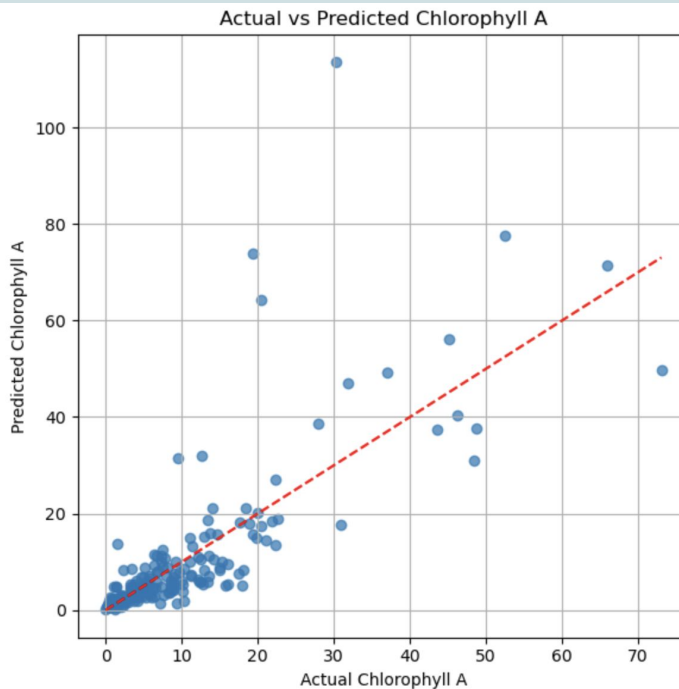
Model Building

- **Support Vector Regressor (SVR)**
 - kernel-based model
 - seeks to find a hyperplane that best fits the data within a specified margin.
 - sensitive to feature scaling and high-dimensional data
 - performed less efficiently
 - computational intensity
 - weak tolerance to noise

Models were implemented using scikit-learn's pipeline architecture to streamline preprocessing, training, and evaluation.

Neural Network

- Predicted Chlorophyll A levels using water quality measurements as input features.
- Pivoted the long-form dataset to wide format so each sample had one row and multiple parameter columns.
- Dropped samples missing Chlorophyll A to ensure a clean and usable target variable.
- Applied mean imputation and feature scaling to prepare data for modeling.
- Built a neural network with two hidden layers (64 and 32 neurons) and dropout layers to reduce overfitting.
- Trained the model for 100 epochs with validation, using the Adam optimizer and MSE loss.
- Achieved low error on the test set, with predictions closely matching actual Chlorophyll A values.
- The model shows strong potential for use in environmental monitoring and early detection of algal blooms.



Mean Absolute Error gives a clear measure of average prediction error in the same units as the target.

- Final **validation MAE** was approximately $\sim X \mu\text{g/L}$, indicating the average absolute prediction error.
- The training and validation MAE curves stayed close, showing that the model is not overfitting and has learned.

Summary of Findings

- **Random Forest Was the Best Performing Model**
 - **R² Score:** 0.6863 ($\approx$ 69% variance explained)
 - **Mean Absolute Error (MAE):** 3.38
 - **Root Mean Squared Error (RMSE):** 6.27
- **XGBoost & SVR Underperformed Relative to Random Forest**
 - **XGBoost** yielded a lower R² of 0.4857 and a higher RMSE of 8.03, suggesting greater individual prediction errors.
 - **Support Vector Regressor (SVR)** performed the worst, with an R² of 0.3673 and the highest RMSE (8.90).
- **Neural Network Successfully Generalized but Was Not Superior**
 - Given the dataset size and nature (tabular), traditional tree-based models remained more interpretable and performant.
- **Important Predictors Identified:**
 - **Total Nitrogen (NTL):** correlation 0.64
 - **Total Phosphorus (PTL):** correlation 0.51
 - Variables like DIN, NH₃, and NO₃ showed lower correlation, and Silica (SiO₂) was nearly irrelevant.
- **Chlorophyll-a Distribution Is Skewed**
 - **CHLA target variable** was **highly right-skewed**, meaning most lakes had low productivity
 - Likely caused underprediction in high-risk lakes

Conclusion

- Limitations
 - **No external validation** limits generalizability beyond the test set.
 - **Temporal dynamics** seasonal effects were not modeled.
 - **High dimensionality** may have affected performance and interpretability.
- Future work
 - Incorporating temporal dynamics: seasonal effects and interannual variability.
 - Leveraging time-series models or recurrent neural networks if longitudinal data becomes available.
 - Include watershed-level data or integrating remote sensing imagery to provides comprehensive view of the drivers of productivity.
 - Enhance model interpretability by using AI techniques like SHAP values for transparency.